



Raj Gaurav Tiwari

Roll No.:230122041

B.Tech - Chemical Science and Technology

Indian Institute Of Technology, Guwahati

+91-9667782966

raj.tiwari@iitg.ac.in

rajt4656@gmail.com

Github | Kaggle

<https://www.linkedin.com/in/rgt2006/>

EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. Major	Indian Institute of Technology, Guwahati	7.95 (Current)	2023-Present
Senior Secondary	CBSE Board	92.4%	2023
Secondary	CBSE Board	96.2%	2021

EXPERIENCE

- Microsoft** May 2026 - Jul. 2026
Software Engineering Intern Hyderabad
 - Built a **dual-detector fraud-detection system** for Microsoft's OneDrive and SharePoint tenants, fusing an **XGBoost** behavioural model (**284 engineered features**) with a **SmartCatch** identity engine using **rapidfuzz** python package.
 - Trained and validated the model to **0.998 PR-AUC / 0.9996 ROC-AUC** with **99.996% precision** on the highest-risk 1% of accounts, then scored **13.4M tenants** at production scale, surfacing **196K (1.46%)** for review.
 - Delivered the end-to-end pipeline as a modular, config-driven **CLI** (Kusto/CSV ingest, feature engineering, gated fusion, **SHAP** explainability) with **60+ automated tests**; the identity detector caught **102K cloned-fraud tenants (52% of flags)** the behavioural model alone missed clones of genuine tenants that were fraudulent.

PROJECTS

- FAULT TOLERANT CONTROL IN QUADROTOR DRONES** Nov. 2024 - Dec. 2024
13th Inter IIT TechMeet, IIT Bombay (Group Project) Github
 - Designed two motor-failure detection and post-failure control algorithms: a **threshold-based** method on sensor-data (**Gyroscope, Accelerometer**) patterns (**0.4–0.6s** latency) and an **MLP** trained on **1000+ simulated scenarios** for real-time status classification, achieving **87.3% accuracy**, **88.4% recall**, and **87.2% F1** at **0.2–0.3s** latency.
 - Developed a custom **PX4 module** for motor fault detection and recovery during mid-flight, integrating **NMPC control logic** into the **control allocator**, validated via simulation across varied controlled environments.
 - Integrated **ROS** and **PX4** via **MicroXRCE Agent** for communication between the **flight controller** and the **GCS**.
- CVALIGN** May 2025 - June 2025
Coding Club, IIT Guwahati Github
 - Designed and developed a full-stack **CV evaluation platform** using **React**, **Tailwind CSS**, **MongoDB** for database and **FastAPI** that automates resume screening using role-based access for admins, recruiters, and hiring managers.
 - Engineered an advanced **Retrieval-Augmented Generation (RAG) pipeline** powered by a local **LLaMA model** to assess CVs based on job descriptions, CGPA, technical skills, achievements, and academic credentials.
 - Delivered ranked CV outputs with **personalized, LLM-based feedback**, helping recruiters make informed decisions.
- PNEUMONIA DETECTOR** Aug. 2024 - Sep. 2024
CLASH OF T-AI-TANS, All India AI Hackathon (Group Project) Github
 - Built a transfer-learning pipeline on **VGG16** to classify chest X-rays as **pneumonia-positive** vs. **normal**, applying hyperparameter tuning, dropout layers, and data augmentation and feature engineering to curb overfitting.
 - Achieved **94.7% accuracy**, **95.4% recall**, **94.6% F1**, and **0.931 AUC-ROC** on the medical imaging dataset.

ACHIEVEMENTS

- JEE Advanced**, Secured an All India Rank of **7393** out of **0.2 million** candidates. 2023
- JEE Mains**, Secured an All India Rank of **12900** out of **1.2 million** candidates. 2023
- 14th Inter IIT TechMeet, IIT Patna**, Secured 2nd position in Drona Aviation PS out of all 23 IITs. 2025
- 13th Inter IIT TechMeet, IIT Bombay**, Secured 6th position in IdeaForge PS out of all 23 IITs. 2024
- Codeforces**, Max rating : **1685**, **Expert**, Handle : **uti_raj**

TECHNICAL SKILLS

- Programming:** C/C++, Python, JavaScript
- Tools/Frameworks:** Keras, TensorFlow, ROS2, OpenCV
- Database Management:** MongoDB, SQL, KQL
- Web Technologies:** ReactJS, NodeJS
- Operating Systems:** Windows, Linux
- Drone Programming:** PX4-Autopilot, MAVSDK

KEY COURSES TAKEN

- Mathematics:** Linear Algebra, Multivariable Calculus, Real Analysis & Complex Analysis.
- Computer Science:** Data Structures & Algorithms, Operating Systems, Object-Oriented Programming.

POSITIONS OF RESPONSIBILITY

- Operations Manager**, Technical Board, IIT Guwahati Apr. 2025 - Apr. 2026
 - Coordinating with **15 clubs** under Technical Board to streamline project execution and manage equipment logistics.
- Project Manager**, Aeromodelling Club, IIT Guwahati Apr. 2025 - Apr. 2026
 - Heading all **UAV projects** for this tenure, providing **technical mentorship** to **50+** juniors, and resource management.